POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5024C | Semester V | Theory | 4 Credits

Composite Materials

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5024C as Composite Materials in Semester V for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently built from Tezpur University’s ME434 Composite Materials course syllabus and the peer-reviewed open-access review by Rajak et al. (2019) on fibre-reinforced polymer composites. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders consistent with adjacent handbook formatting. Manufacturing, testing, chemical handling, machining and disposal must follow the laboratory’s risk assessment, SDS information, equipment manual and competent-supervision requirements.

Verified source note: The verified sources support composite classification, matrices and reinforcement forms, fabrication routes, basic mechanics, testing, defects, applications and recent developments. The handbook is organised into four learner-friendly modules without claiming an official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Composite Materials introduces engineered combinations of matrix and reinforcement that deliver tailored strength, stiffness, weight, durability and functional performance. The course connects constituent selection, processing, mechanics, testing, defects and responsible application.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Engineering materials, basic mechanics, manufacturing processes, safe workshop/laboratory practice and the ability to interpret simple stress–strain data.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the need, classification, constituents and applications of composite materials.
2. Identify reinforcement forms, matrix systems and selection requirements.
3. Explain common composite-manufacturing routes and their process-quality links.
4. Explain load transfer, mechanical testing, failure modes and emerging composite technologies.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Classify composites and relate constituent choice to required properties and applications.	Understand
CO2	Explain reinforcements, matrices and selection of composite constituents.	Understand
CO3	Explain principal composite fabrication routes, process variables and quality concerns.	Understand
CO4	Explain composite load response, testing, failure mechanisms and recent developments.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Composite Fundamentals and Applications	Need for composites; matrix and reinforcement roles; classifications; property tailoring; advantages, limitations and applications.	Approx. 14 hours	CO1	Module 1
2	Reinforcements, Matrices and Selection	Fibre, particle and whisker reinforcement; fabric architectures; polymer, metal, ceramic and carbon-based matrices; constituent selection and interface awareness.	Approx. 15 hours	CO2	Module 2
3	Fabrication, Quality and Process Choice	Hand lay-up, spray-up, compression moulding, RTM, pultrusion, filament winding, injection moulding and selected metal-matrix processing; defects and quality controls.	Approx. 16 hours	CO3	Module 3
4	Mechanics, Testing and Emerging Composites	Iso-strain/iso-stress ideas, anisotropy, fibre orientation, stress–strain response, tensile/flexural/interfacial tests, failure modes, toughening and sustainable/advanced composites.	Approx. 15 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Composite Fundamentals and Applications

Need for composites; matrix and reinforcement roles; classifications; property tailoring; advantages, limitations and applications.

CO1 · Approx. 14 hours

Reinforcements, Matrices and Selection

Fibre, particle and whisker reinforcement; fabric architectures; polymer, metal, ceramic and carbon-based matrices; constituent selection and interface awareness.

CO2 · Approx. 15 hours

Fabrication, Quality and Process Choice

Hand lay-up, spray-up, compression moulding, RTM, pultrusion, filament winding, injection moulding and selected metal-matrix processing; defects and quality controls.

CO3 · Approx. 16 hours

Mechanics, Testing and Emerging Composites

Iso-strain/iso-stress ideas, anisotropy, fibre orientation, stress–strain response, tensile/flexural/interfacial tests, failure modes, toughening and sustainable/advanced composites.

CO4 · Approx. 15 hours

MODULE 1

Composite Fundamentals and Applications

Need for composites; matrix and reinforcement roles; classifications; property tailoring; advantages, limitations and applications.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

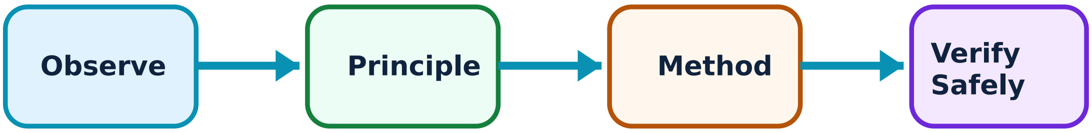
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Composite Fundamentals and Applications: identify the system, state its principle, follow the correct method and verify the result safely.

1. What makes a composite?−

A composite combines two or more distinct constituents so that the matrix binds, transfers load and protects the reinforcement while the reinforcement contributes strength, stiffness or other performance. The interface is important because load must transfer between phases.

Study and application method

For a proposed product, name the matrix, reinforcement, likely interface role and the performance target such as lower mass or better corrosion resistance.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a proposed product, name the matrix, reinforcement, likely interface role and the performance target such as lower mass or better corrosion resistance.

> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

> **Malayalam support:** □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

> **Common mistake / precaution:** A high specific property does not automatically make a material suitable; cost, joining, damage tolerance, recycling, environment and service temperature also matter.

2. Classification and applications−

Composites may be classified by matrix as polymer-matrix (PMC), metal-matrix (MMC) or ceramic-matrix (CMC), and by reinforcement as fibre, particle, whisker, laminate or hybrid. Applications include automotive, aerospace, marine, civil, biomedical, renewable-energy and sports products.

Study and application method

Build a classification tree, then justify one material-system/application pairing using service loads and environmental conditions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Application examples are not design approval; real components require validated material data, codes and design review.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Reinforcements, Matrices and Selection

Fibre, particle and whisker reinforcement; fabric architectures; polymer, metal, ceramic and carbon-based matrices; constituent selection and interface awareness.

Approx. 15 hours | CO2 | Understand · Apply

Learning goals

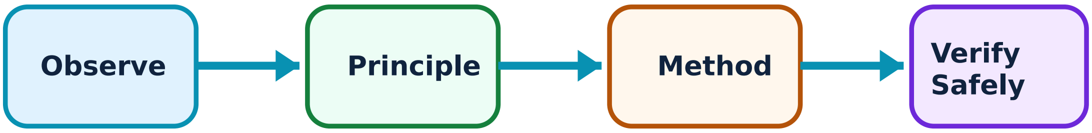
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Reinforcements, Matrices and Selection: identify the system, state its principle, follow the correct method and verify the result safely.

1. Reinforcement materials and forms−

Glass, carbon, aramid, basalt and natural fibres, along with particles, whiskers and nano-fillers, can reinforce a matrix. Performance depends not only on material type but also on fibre length, volume fraction, orientation, weave and dispersion.

Study and application method

Compare two reinforcement options against a stated requirement using specific strength/stiffness, cost, moisture or temperature resistance, processing compatibility and end-of-life concerns.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare two reinforcement options against a stated requirement using specific strength/stiffness, cost, moisture or temperature resistance, processing compatibility and end-of-life concerns.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Fine fibres, dust and nano-materials can create inhalation or skin hazards; use the specified controls and do not handle loose material without authorisation.

2. Matrix systems and constituent selection−

Polymer matrices are widely used for low density and processability; metal and ceramic matrices support higher-temperature or specialised service. Selection also considers wetting, interfacial bonding, curing/processing route, corrosion, thermal expansion and recyclability.

Study and application method

Use a requirements table: service temperature, load direction, fluid exposure, production volume, allowable cost and disposal route; then state a defensible provisional material choice.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a requirements table: service temperature, load direction, fluid exposure, production volume, allowable cost and disposal route; then state a defensible provisional material choice.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not treat a supplier data sheet as universal performance evidence: curing, fibre architecture, voids and test direction affect the actual laminate.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Fabrication, Quality and Process Choice

Hand lay-up, spray-up, compression moulding, RTM, pultrusion, filament winding, injection moulding and selected metal-matrix processing; defects and quality controls.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

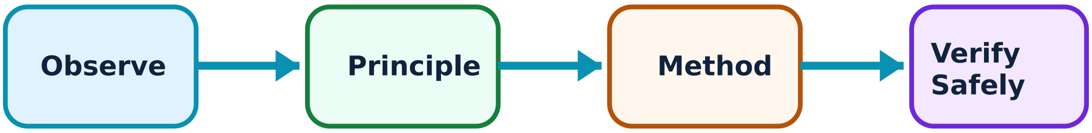
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Fabrication, Quality and Process Choice: identify the system, state its principle, follow the correct method and verify the result safely.

1. Polymer-composite manufacturing routes−

Open-mould routes such as hand lay-up and spray-up suit simpler shapes; closed or automated routes including resin-transfer moulding, compression moulding, pultrusion, filament winding and injection moulding offer different productivity, repeatability and geometry capabilities.

Study and application method

Map the material flow for one route: preparation, reinforcement placement, resin introduction or consolidation, cure/cooling, demoulding and inspection. State why the route fits the part geometry and production quantity.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map the material flow for one route: preparation, reinforcement placement, resin introduction or consolidation, cure/cooling, demoulding and inspection. State why the route fits the part geometry and production quantity.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Resins, catalysts, solvents, heat and rotating equipment present chemical, fire and mechanical hazards. Use only the authorised procedure, PPE, ventilation and waste route.

2. Process variables, defects and quality−

Fibre misalignment, poor wet-out, voids, resin-rich zones, incomplete cure, delamination and fibre/matrix debonding can reduce expected performance. Material preparation, process parameters and inspection must be controlled together.

Study and application method

Create a cause–effect table connecting a defect, likely process cause, observable symptom, prevention step and appropriate non-destructive or destructive evaluation route.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a cause–effect table connecting a defect, likely process cause, observable symptom, prevention step and appropriate non-destructive or destructive evaluation route.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Never infer safety or structural acceptability from appearance alone; acceptance criteria and inspection methods are product- and standard-specific.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Mechanics, Testing and Emerging Composites

Iso-strain/iso-stress ideas, anisotropy, fibre orientation, stress–strain response, tensile/flexural/interfacial tests, failure modes, toughening and sustainable/advanced composites.

Approx. 15 hours | **CO4** | **Understand · Apply**

Learning goals

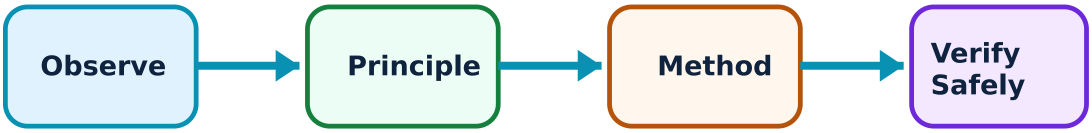
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Mechanics, Testing and Emerging Composites: identify the system, state its principle, follow the correct method and verify the result safely.

1. Load transfer and mechanical response−

Continuous aligned fibres can carry load effectively along their direction. Iso-strain and iso-stress conditions are simplified models for understanding constituent response; real laminates are anisotropic and depend strongly on fibre orientation, interface and stacking sequence.

Study and application method

Sketch a load direction relative to fibres, predict whether the response is fibre-dominated or matrix/interface-dominated, and state the assumptions before applying a simple rule-of-mixtures estimate.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a load direction relative to fibres, predict whether the response is fibre-dominated or matrix/interface-dominated, and state the assumptions before applying a simple rule-of-mixtures estimate.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Do not apply a simple mixture rule to a laminate without stating orientation, fibre fraction, void content, interface quality and test direction.

2. Testing, failure and recent developments−

Tensile, compression, flexural, shear and interfacial tests help characterise composites. Fibre fracture, matrix cracking, debonding, pull-out and delamination are important failure mechanisms. Current directions include bio-composites, nano-composites, hybrid systems, self-healing concepts and improved circularity.

Study and application method

For a test result, record specimen architecture, conditioning, direction, loading mode, units and failure observation; then connect the observed failure to a plausible design or process improvement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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> **Common mistake / precaution:** Testing machines and damaged composites can release energy, sharp fragments or dust. Follow approved guarding, specimen handling and disposal procedures.

> **Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

> State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Composite Fundamentals and Applications

Use the concept flow to revise observation, principle, method and verification.

Module 2: Reinforcements, Matrices and Selection

Use the concept flow to revise observation, principle, method and verification.

Module 3: Fabrication, Quality and Process Choice

Use the concept flow to revise observation, principle, method and verification.

Module 4: Mechanics, Testing and Emerging Composites

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a constituent-selection matrix for an automotive interior panel, a wind-turbine component and a corrosion-exposed cover.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw and annotate the material and process flow for hand lay-up, RTM, pultrusion or filament winding.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Interpret supplied tensile and three-point-bend data, noting direction, units, assumptions and likely failure mechanism.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Conduct a supervised defect-identification exercise using photographs or safe sample coupons; propose preventive quality controls.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Debate a material-choice case that balances performance, worker safety, cost, repairability and end-of-life treatment.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Composite Fundamentals and Applications

Explain What makes a composite? and state one correct method, application or precaution.−

A composite combines two or more distinct constituents so that the matrix binds, transfers load and protects the reinforcement while the reinforcement contributes strength, stiffness or other performance. The interface is important because load must transfer between phases.

Answer framework: For a proposed product, name the matrix, reinforcement, likely interface role and the performance target such as lower mass or better corrosion resistance.

Important point: A high specific property does not automatically make a material suitable; cost, joining, damage tolerance, recycling, environment and service temperature also matter.

Explain Classification and applications and state one correct method, application or precaution.−

Composites may be classified by matrix as polymer-matrix (PMC), metal-matrix (MMC) or ceramic-matrix (CMC), and by reinforcement as fibre, particle, whisker, laminate or hybrid. Applications include automotive, aerospace, marine, civil, biomedical, renewable-energy and sports products.

Answer framework: Build a classification tree, then justify one material-system/application pairing using service loads and environmental conditions.

Important point: Application examples are not design approval; real components require validated material data, codes and design review.

Module 2 — Reinforcements, Matrices and Selection

Explain Reinforcement materials and forms and state one correct method, application or precaution.−

Glass, carbon, aramid, basalt and natural fibres, along with particles, whiskers and nano-fillers, can reinforce a matrix. Performance depends not only on material type but also on fibre length, volume fraction, orientation, weave and dispersion.

Answer framework: Compare two reinforcement options against a stated requirement using specific strength/stiffness, cost, moisture or temperature resistance, processing compatibility and end-of-life concerns.

Important point: Fine fibres, dust and nano-materials can create inhalation or skin hazards; use the specified controls and do not handle loose material without authorisation.

Explain Matrix systems and constituent selection and state one correct method, application or precaution.−

Polymer matrices are widely used for low density and processability; metal and ceramic matrices support higher-temperature or specialised service. Selection also considers wetting, interfacial bonding, curing/processing route, corrosion, thermal expansion and recyclability.

Answer framework: Use a requirements table: service temperature, load direction, fluid exposure, production volume, allowable cost and disposal route; then state a defensible provisional material choice.

Important point: Do not treat a supplier data sheet as universal performance evidence: curing, fibre architecture, voids and test direction affect the actual laminate.

Module 3 — Fabrication, Quality and Process Choice

Explain Polymer-composite manufacturing routes and state one correct method, application or precaution.−

Open-mould routes such as hand lay-up and spray-up suit simpler shapes; closed or automated routes including resin-transfer moulding, compression moulding, pultrusion, filament winding and injection moulding offer different productivity, repeatability and geometry capabilities.

Answer framework: Map the material flow for one route: preparation, reinforcement placement, resin introduction or consolidation, cure/cooling, demoulding and inspection. State why the route fits the part geometry and production quantity.

Important point: Resins, catalysts, solvents, heat and rotating equipment present chemical, fire and mechanical hazards. Use only the authorised procedure, PPE, ventilation and waste route.

Explain Process variables, defects and quality and state one correct method, application or precaution.−

Fibre misalignment, poor wet-out, voids, resin-rich zones, incomplete cure, delamination and fibre/matrix debonding can reduce expected performance. Material preparation, process parameters and inspection must be controlled together.

Answer framework: Create a cause–effect table connecting a defect, likely process cause, observable symptom, prevention step and appropriate non-destructive or destructive evaluation route.

Important point: Never infer safety or structural acceptability from appearance alone; acceptance criteria and inspection methods are product- and standard-specific.

Module 4 — Mechanics, Testing and Emerging Composites

Explain Load transfer and mechanical response and state one correct method, application or precaution.−

Continuous aligned fibres can carry load effectively along their direction. Iso-strain and iso-stress conditions are simplified models for understanding constituent response; real laminates are anisotropic and depend strongly on fibre orientation, interface and stacking sequence.

Answer framework: Sketch a load direction relative to fibres, predict whether the response is fibre-dominated or matrix/interface-dominated, and state the assumptions before applying a simple rule-of-mixtures estimate.

Important point: Do not apply a simple mixture rule to a laminate without stating orientation, fibre fraction, void content, interface quality and test direction.

Explain Testing, failure and recent developments and state one correct method, application or precaution.−

Tensile, compression, flexural, shear and interfacial tests help characterise composites. Fibre fracture, matrix cracking, debonding, pull-out and delamination are important failure mechanisms. Current directions include bio-composites, nano-composites, hybrid systems, self-healing concepts and improved circularity.

Answer framework: For a test result, record specimen architecture, conditioning, direction, loading mode, units and failure observation; then connect the observed failure to a plausible design or process improvement.

Important point: Testing machines and damaged composites can release energy, sharp fragments or dust. Follow approved guarding, specimen handling and disposal procedures.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Composite Fundamentals and Applications.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Reinforcements, Matrices and Selection.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Fabrication, Quality and Process Choice.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Mechanics, Testing and Emerging Composites.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Need for composites; matrix and reinforcement roles; classifications; property tailoring; advantages, limitations and

applications..
2. Develop a complete answer or practical plan for: Fibre, particle and whisker reinforcement; fabric architectures; polymer, metal, ceramic and carbon-based matrices; constituent selection and interface awareness..
3. Develop a complete answer or practical plan for: Hand lay-up, spray-up, compression moulding, RTM, pultrusion, filament winding, injection moulding and selected metal-matrix processing; defects and quality controls..
4. Develop a complete answer or practical plan for: Iso-strain/iso-stress ideas, anisotropy, fibre orientation, stress–strain response, tensile/flexural/interfacial tests, failure modes, toughening and sustainable/advanced composites..

LAST-MILE REVISION

Quick Revision

Module 1: Composite Fundamentals and Applications

Need for composites; matrix and reinforcement roles; classifications; property tailoring; advantages, limitations and applications.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Reinforcements, Matrices and Selection

Fibre, particle and whisker reinforcement; fabric architectures; polymer, metal, ceramic and carbon-based matrices; constituent selection and interface awareness.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Fabrication, Quality and Process Choice

Hand lay-up, spray-up, compression moulding, RTM, pultrusion, filament winding, injection moulding and selected metal-matrix processing; defects and quality controls.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Mechanics, Testing and Emerging Composites

Iso-strain/iso-stress ideas, anisotropy, fibre orientation, stress–strain response, tensile/flexural/interfacial tests, failure modes, toughening and sustainable/advanced composites.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
What makes a composite?	A composite combines two or more distinct constituents so that the matrix binds, transfers load and protects the reinforcement while the reinforcement contributes strength, stiffness or other performance.
Classification and applications	Composites may be classified by matrix as polymer-matrix (PMC), metal-matrix (MMC) or ceramic-matrix (CMC), and by reinforcement as fibre, particle, whisker, laminate or hybrid.
Reinforcement materials and forms	Glass, carbon, aramid, basalt and natural fibres, along with particles, whiskers and nano-fillers, can reinforce a matrix.
Matrix systems and constituent selection	Polymer matrices are widely used for low density and processability; metal and ceramic matrices support higher-temperature or specialised service.
Polymer-composite manufacturing routes	Open-mould routes such as hand lay-up and spray-up suit simpler shapes; closed or automated routes including resin-transfer moulding, compression moulding, pultrusion, filament winding and injection moulding offer different productivity, repeatability and geometry capabilities.
Process variables, defects and quality	Fibre misalignment, poor wet-out, voids, resin-rich zones, incomplete cure, delamination and fibre/matrix debonding can reduce expected performance.
Load transfer and mechanical response	Continuous aligned fibres can carry load effectively along their direction.
Testing, failure and recent developments	Tensile, compression, flexural, shear and interfacial tests help characterise composites.

LEARNING RESOURCES

References

Recommended composite-materials references

1. K. K. Chawla, Composite Materials.
2. F. L. Matthews and R. D. Rawlings, Composite Materials: Engineering and Science.
3. A. B. Strong, Fundamentals of Composite Manufacturing.
4. P. K. Mallick, Composite Materials: Technology, Processes and Applications.

Approved public composite-materials sources

- https://www.tezu.ernet.in/dmech/student_NB/course_list_plan_s22/ME434.pdf
- https://pmc.ncbi.nlm.nih.gov/articles/PMC6835861/

These sources provide the verified learning basis while official Course 5024C detail is unavailable; they do not replace authorised laboratory or industrial procedures.